

Distributed Systems — Homework 3

MapReduce and gRPC: Design, Analysis and Evaluation

1 Introduction

This report presents four distributed programs: two built on the MapReduce model and two on gRPC. Section 2 implements distributed matrix multiplication. Section 3 implements the Homework 2 log-analytics problem as a batch MapReduce job, and Section 4 implements the same analytics as a real-time streaming system, so that the two paradigms can be compared on identical work. Section 5 implements a food ordering service. Section 6 describes the execution environment and the measurement method; Section 7 draws the comparisons together.

The emphasis throughout is on *why* each design was chosen and what the measurements reveal about it. All figures reported here were obtained on the RCE cluster and are reproduced directly from the recorded result files.

Execution model for MapReduce. The assignment specifies Apache Hadoop 3.3.6 on YARN. This is not available on RCE: the shared HDFS is read-only for a student account (its root is owned by `hdfs:supergroup` with mode `drwxr-xr-x`) and no ResourceManager is configured, so no YARN application can be submitted. The course has confirmed a known issue with the Hadoop environment and that a Slurm-based script is acceptable in the interim. Both MapReduce programs are therefore driven by shell scripts that reproduce the Hadoop Streaming execution flow exactly:

split → map (parallel) → sort → combine → sort → reduce

The mapper, combiner and reducer are standalone executables communicating only through standard input and standard output, which is precisely the Hadoop Streaming contract; they would run unmodified under a working Hadoop installation.

2 Distributed Matrix Multiplication

2.1 Problem and approach

Given A of dimension $m \times n$ and B of dimension $n \times p$, compute $C = A \times B$ by the **Row–Row method**, in which each row of C is a weighted sum of the rows of B :

$$c_i = a_{i,1}B_{1,:} + a_{i,2}B_{2,:} + \cdots + a_{i,n}B_{n,:}$$

This formulation is attractive for MapReduce because each row of C depends on exactly one row of A and on all of B . If B is replicated to every mapper, a mapper can compute a complete contribution to c_i from row i alone, with no communication between mappers. The row index is then a natural reduce key.

2.2 Data distribution

A is stored row-by-row and split across mappers; B is **replicated into the local memory of every mapper** before processing begins. This asymmetry is deliberate: A is the large operand and is partitioned, while B is broadcast so that no mapper ever needs data held by another. Each input row of A carries its own global row index, so a mapper can label its output correctly without knowing which split it received.

2.3 Key–value schema

Stage	Key	Value
Map output	row index i , zero-padded	partial vector $a_{ik} \cdot B_{k,:}$, one per k
Combine	row index i	element-wise sum of that node’s partial vectors for i
Reduce out	—	the completed row c_i , emitted in ascending i

Keys are zero-padded so that lexicographic sorting, which is what the shuffle performs, agrees with numeric ordering. The combiner consumes and produces the same format as the mapper, so the reducer cannot distinguish a run in which the combiner executed from one in which it did not — the property a combiner must satisfy. It is valid here because vector addition is associative and commutative, so partial sums may be grouped arbitrarily.

2.4 Complexity

A mapper emits n partial vectors of length p for each of its rows, so a split of m' rows produces $O(m'np)$ intermediate values. The combiner reduces this to $O(m'p)$ before the network shuffle — a factor of n — which is the single most important property of the design, since intermediate volume rather than input size governs the cost. The reducer performs $O(mp)$ additions in total.

2.5 Correctness

Every result is checked against a direct triple-loop multiplication of the same matrices. Nine configurations were executed distributed across four nodes, chosen to cover the cases the problem statement identifies as significant: the two worked examples from the assignment; the edge cases $m = 1$ and $n = 1$; inputs whose row count is and is not divisible by the number of mappers; and skewed shapes in both directions. All nine produced results identical to the reference computation.

Configuration	A	B	m divisible by 4
Assignment Example 1	3×2	2×3	no
Assignment Example 2	4×2	2×2	yes
Edge case, single row	1×3	3×2	no
Edge case, single column	3×1	1×3	no
Baseline	100×50	50×50	yes
Square	1000×100	100×1000	yes
Square, indivisible	1001×100	100×1000	no
Tall, $m \gg n$	5000×10	10×10	yes
Wide, $n \gg m$	10×500	500×10	no

2.6 Performance

Four nodes, four map tasks. Times in seconds.

Case	A	B	div.	Map	Shuffle	Combine	Reduce	Total
p1_small	100x50	50x50	yes	0.22	0.17	0.18	0.02	0.60
p1_medium	500x50	50x50	yes	0.31	0.22	0.24	0.03	0.82
p1_large	2000x50	50x50	yes	0.76	0.38	0.46	0.09	1.72
p1_square	1000x100	100x1000	yes	11.05	12.30	5.53	0.69	29.63
p1_square_uneven	1001x100	100x1000	no	11.07	13.11	5.54	0.69	30.47
p1_tall	5000x10	10x10	yes	0.27	0.19	0.23	0.06	0.77
p1_wide	10x500	500x10	no	0.18	0.17	0.19	0.02	0.57

Intermediate volume dominates, not input size. The two square cases take roughly 30 s while 2000×50 takes 1.7 s, although their inputs are of comparable size. The difference follows directly from the $O(mnp)$ bound: with $n = 100$ and $p = 1000$ each row produces 100 vectors of 1000 values, some 400 MB of intermediate text, and the map and shuffle stages carry that cost.

Indivisible splits cost almost nothing. 1000 rows and 1001 rows differ by 0.85 s out of 30 s; the single extra row is handled by one node with no special case in the code.

The combiner is what keeps the global merge cheap. Although the intermediate data reaches hundreds of megabytes, the second sort takes 0.05 s, because each node has already collapsed its partial vectors to one row each before anything crosses the network. This is the $O(n)$ reduction of the complexity analysis, visible directly in the timings.

Shape determines utilisation. The tall case distributes 5000 rows over four nodes and finishes in 0.77 s; the wide case has only 10 rows to distribute, so most nodes are idle and the run is limited by a single mapper regardless of cluster size.

3 Batch Log Analytics with MapReduce

3.1 Problem

The Homework 2 analytics over a server log: total, successful and failed request counts; counts by status class; minimum, maximum and mean response time; total bytes transferred; the busiest 60-second interval; and the Top- K servers and endpoints. Implemented in C++ as mapper, combiner and reducer executables.

3.2 The property that makes the job decomposable

Every quantity retained is a **count**, a **sum**, a **minimum** or a **maximum**. Each of these is associative and commutative, so a partial result computed over any subset of the input can be combined with any other, in any order, to give exactly the result that would have been obtained from the whole. This is the precondition for splitting the work at all, and it dictates two design decisions:

- **Averages are never stored.** A mean is not mergeable — the mean of two means is not the mean of the union unless the counts are equal. The sum and the count are stored instead, and the division is performed once, when the final answer is printed.
- **Response times are stored as integers**, in units of 10^{-6} ms. Floating point addition is not associative, so a floating-point total would depend on the order in which mappers happened to be merged, and that order varies between runs. Integers make the result reproducible and exactly equal to the sequential program's.

3.3 In-mapper combining

A mapper does not emit one record per input line. It accumulates its entire split in memory and emits one pair per *distinct key* it observed:

Key	Meaning	Value
K	the Top- K size	taken from the dataset header
G	split-wide totals	counts, byte total, time sum, minimum, maximum
S: id	one server	request count, summed response time
E: id	one endpoint	request count, summed bytes
I: t	one 60-second interval	request count

The consequence is that intermediate data is bounded by the *key space* — the number of distinct servers, endpoints and intervals — and not by the number of records. On the ten-million-record dataset the shuffle is **278× smaller than the input**. Since the sort is the expensive part of a MapReduce job, this is the decision that determines the program’s performance.

3.4 Why a single reducer

Most of the analytics could be computed by any number of reducers and combined afterwards. The Top- K lists could not: identifying the ten busiest servers requires the totals of *all* servers, and a server ranked eleventh on every reducer may still rank first overall. A single reducer therefore observes the complete key space. This is affordable only because in-mapper combining has already reduced the data it must read to a few megabytes.

3.5 Correctness

The requirement is stricter than numerical agreement: the output must match the Homework 2 sequential program byte for byte. Thirty-four checks were performed, covering the worked example; five edge cases (empty input, a single record, ties in the Top- K ordering, $K = 0$, and response times as small as 10^{-6} ms); and the same input divided into 1, 2, 4 and 8 splits, since a correct decomposition must give an identical answer however the input is divided. All thirty-four matched, as did all 24 runs of the performance sweep described below.

3.6 Performance

Map tasks placed one per node across four nodes. Times in seconds.

Map tasks	100k records	1M records	10M records	10M: map stage	Speedup
1	0.342	1.701	16.127	15.535	1.00×
2	0.318	1.095	9.383	8.872	1.72×
4	0.296	0.696	4.922	4.453	3.28×
8	0.599	1.026	5.182	4.469	3.11×

The map stage is the computation. It accounts for 96% of the single-task run. Measured separately on a compute node, reading the 397 MB input with `cat` takes 0.26 s while the mapper takes 13.26 s, of which 13.16 s is user CPU time. The work is parsing ten million lines of text, not transferring them, and the program is therefore compute-bound and scales with available cores.

Parallelism is limited by physical cores, not by the algorithm. Speedup tracks the map stage closely up to four tasks (3.28× on four nodes). At eight tasks the map stage does not improve

at all, because the scheduler places the second task on some nodes onto a sibling hyperthread of a core already in use; two processes sharing one physical core take twice as long each. The additional tasks still enlarge the sort and combine stages, so the total time rises slightly. This is a property of the allocation, not of the decomposition.

The serial tail is present but small. The final gather, sort and reduce do not shrink as mappers are added; they grow from 0.22s to 0.28s, because more mappers emit more partial rows for the same keys. This is the serial fraction of Amdahl’s law, and at under 6% of the four-task run it is not the binding constraint here.

Small inputs should not be distributed. At 100k records the four-task run is barely faster than the single-task run and the eight-task run is clearly slower: process startup across nodes exceeds the work being parallelised. MapReduce carries a fixed overhead that is only amortised from roughly a million records upward.

3.7 Comparison with MPI

The same analytics were implemented with MPI, splitting the input by byte range and combining with `MPI_Reduce` (`MPI_SUM` for counts and sums, `MPI_MIN` and `MPI_MAX` for the extremes). Both systems were given equal core counts. MPI runs all ranks on one node, since Open MPI cannot launch across nodes in this environment; MapReduce distributes its tasks over four. All 24 MPI runs reproduced the sequential output.

Processes	MPI	MapReduce	Ratio	MPI speedup	MapReduce speedup
1	7.611	16.127	2.12×	1.00×	1.00×
2	4.437	9.383	2.11×	1.72×	1.72×
4	2.225	4.922	2.21×	3.42×	3.28×
8	2.194	5.182	2.36×	3.47×	3.11×

MPI is between 1.4 and 2.4 times faster at every size, and the ratio is stable at about 2.1× on the largest dataset. This is the expected outcome: MPI performs no shuffle, no sort, no serialisation to text and no process creation per stage. It holds its state in memory and reduces once at the end.

The two scale almost identically — 3.42× and 3.28× respectively at four processes — so the parallel decomposition is equally effective in both. The difference is entirely in the constant factor, which makes it worth identifying where that constant comes from.

3.8 Where the constant factor comes from

The two programs differ in how they represent state. Measured on the ten-million-record dataset:

Program	State representation	Peak memory	Time
Sequential (HW2)	two passes; all records buffered; dense arrays	395 MB	4.65 s
MPI, one rank	as above, with a byte-range split	403 MB	6.31 s
Mapper (this work)	one streaming pass; sparse maps	10.1 MB	13.00 s

The Homework 2 design pre-scans the input to determine the range of server and endpoint identifiers, allocates dense arrays, and then indexes them directly: an array subscript per record, which is as fast as this computation can be. The cost is that it must hold all ten million records in memory first.

The mapper cannot do this. A mapper sees only its own split and therefore cannot know the global identifier ranges in advance, so it counts in a single streaming pass into sparse associative containers, discarding each record once counted. The price is a tree lookup per record rather than an array subscript.

The result is a **39× reduction in memory for a factor of about 2.8 in speed**. This is not a defect in either program but a genuine engineering trade, and the memory property is what permits the MapReduce decomposition in the first place: a design requiring 395 MB per task and a prior pass over the whole input cannot be split across mappers that each see a fraction of the data. A single-pass MPI implementation using sparse state would combine both advantages and would be expected to outperform everything measured here.

4 Real-Time Log Analytics with gRPC

4.1 Problem and architecture

The same analytics, but with the data treated as a continuous stream rather than a complete batch, and with the current results queryable at any time during ingestion.

```
stream_client --StreamRecords--> +-- Coordinator --+ --Process()--> Worker 0
(replays the dataset)           | router           | --Process()--> Worker 1
                                | bounded queues | --Process()--> Worker 2
dashboard --WatchAnalytics---> | adds workers up| <--GetStats()-- (each counts
query_client --GetAnalytics---> +-----+ into its own)
```

The coordinator performs three roles: it accepts record streams, routes each batch to a worker, and answers queries by combining the workers’ partial results. Workers count records into their own state and never communicate with one another or with clients. The mergeability property established in Section 3 is what makes this possible: the global answer is the combination of the workers’ states, computed on demand.

4.2 Interface design

RPC	Kind	Rationale
StreamRecords	client streaming	One long-lived call carries an entire stream: a header, then any number of batches. Using one call per stream rather than per batch avoids repeated connection and header costs, and gRPC’s flow control then supplies backpressure without additional mechanism.
GetAnalytics	unary	The current analytics, answered while ingestion continues.
WatchAnalytics	server streaming	A snapshot pushed at a fixed interval, so that N dashboards cost one combination per refresh rather than N .
Worker.Process	client streaming	All batches for a worker travel on a single persistent call.
Worker.GetStats	unary	Returns the worker’s complete running totals.

Idempotent aggregation. Each worker maintains its own running totals and always returns all of them; the coordinator *replaces* that worker’s contribution rather than accumulating it. Repeating a request is therefore harmless, which removes the need for sequence numbers, acknowledgements or duplicate detection. An earlier version transmitted numbered deltas that the coordinator had

to acknowledge, with recovery rules for lost replies; replacing rather than accumulating made that protocol unnecessary and removed a substantial amount of code.

Columnar batches. A batch is encoded as one packed array per field rather than as repeated sub-messages, so a 1000-record batch is seven packed arrays instead of a thousand small messages. Protocol Buffers decodes packed scalars in native code, and the worker iterates the columns directly.

4.3 Concurrency and flow control

Each worker has a **bounded** queue. When a worker falls behind, its queue fills, the enqueue operation blocks, the ingestion handler stops reading from the stream, and gRPC’s flow control propagates the pause to the client. Memory therefore remains bounded regardless of how fast the source sends. A worker’s state is guarded by a single lock held for a whole batch, so a concurrent query can never observe a partially applied batch.

4.4 Correctness

Once the stream has been fully processed, the analytics must equal the sequential program’s output exactly. Seventy-six checks were performed: one to three workers combined with three distribution strategies and several batch sizes; two sources streaming concurrently; queries issued while ingestion was still in progress; and K supplied at query time rather than taken from the dataset header. All matched, as did all 36 runs of the performance sweep.

4.5 Performance

One million records, one process per node, coordinator, workers and clients all on separate machines.

Effect of worker count

Workers	records/s	Speedup
1	799,329	1.00×
2	1,555,611	1.95×
4	2,999,449	3.75×

Effect of distribution strategy

Strategy	records/s	Relative
round_robin	2,999,449	1.00×
least_loaded	2,285,566	0.76×
hash_server	220,454	0.07×

Effect of message granularity

Records per message	Messages per million	records/s	Relative
1	1,000,000	10,126	1×
10	100,000	99,549	10×
100	10,000	844,397	83×
1,000	1,000	2,999,449	296×
10,000	100	3,104,559	307×

Effect of concurrent queries. Each client issues queries in a closed loop with no pause between them, which is the most demanding pattern possible.

Query clients	Ingest (rec/s)	Change	Queries/s	p50 (ms)	p99 (ms)
none	2,999,449	—	—	—	—
1	2,371,485	-21%	49.8	21.317	25.297
4	807,953	-73%	146.1	30.302	41.806
16	204,572	-93%	153.5	113.811	181.138

Scaling is near linear to four workers ($3.75\times$), because workers share no state and require no coordination. The shortfall from a perfect $4\times$ is attributable to the coordinator, through which every record must pass.

Message granularity is the dominant parameter, worth $296\times$ between one record per message and one thousand. Up to a batch of 100 the throughput rises almost exactly tenfold for each tenfold increase in batch size, which identifies the cost as *per message* rather than per record: framing, a handler iteration, a queue transfer and a forwarding call are each paid once per message irrespective of its contents. Beyond 1000 the curve flattens, since per-message overhead has been amortised and per-record work dominates. A batch of 1000 is therefore the appropriate default: it lies at the start of the plateau, and at these rates a batch waits about one millisecond to fill, so the analytics remain effectively current.

Partitioning by key is expensive and unnecessary here. The `hash_server` strategy is roughly fourteen times slower than round robin because it is the only strategy that examines every *record* rather than every *batch*: the coordinator must disassemble each batch and re-encode one batch per worker, which is $O(\text{records})$ work in the single process that is already the bottleneck. Key affinity would be required if a worker had to hold all state for a given key, but every analytic here is mergeable, so it is not.

Load-aware routing did not pay off. `least_loaded` is slower than round robin, contrary to its intent: it inspects every worker queue for every batch, and since the workers are near-identical it pays that cost without ever finding a meaningfully shorter queue. It would be justified only with heterogeneous or contended workers.

Queries are served throughout ingestion, and they are not free. A single closed-loop client reduces ingest throughput by 21%, four by 73% and sixteen by 93%, while the rate of answers saturates near 150 per second. Each query requires the coordinator to contact every worker and combine their states, and this occurs in the same single-threaded process that routes records. Beyond roughly four concurrent closed-loop clients the coordinator is saturated and further clients merely queue. **This is the principal limitation of the architecture.** Two remedies follow directly from the diagnosis: partition the coordinator so that routing and query service are not serialised through one process, or answer queries from a snapshot refreshed on a timer, trading a bounded amount of staleness for throughput.

5 Food Ordering System with gRPC

5.1 Structure

A central server maintains a fixed catalogue of restaurants and the state of all orders. Customer clients and restaurant clients are separate processes, run on separate nodes, and interact only through the server.

The service exposes `ListRestaurants`, `PlaceOrder`, `GetOrderStatus`, `UpdateOrderStatus` and `SubscribeToOrderUpdates`, together with a helper, `ListRestaurantOrders`, which allows a restaurant client to retrieve its own pending orders without maintaining client-side state. `SubscribeToOrderUpdates` is a server-streaming call: a customer subscribes to an order and the server pushes each status change as it occurs.

5.2 Order lifecycle

An order progresses through `PLACED` $\rightarrow$ `ACCEPTED` $\rightarrow$ `PREPARING` $\rightarrow$ `READY`, and may instead be cancelled from `PLACED`. The permitted transitions are held in an explicit table and any transition absent from it is refused, so an order cannot move backwards and cannot be cancelled once a restaurant has accepted it. Encoding the state machine as data rather than as scattered conditionals

means the validity rule is stated in one place.

5.3 Concurrency and error handling

All shared state is protected by a single lock. Notifications to subscribers are dispatched *outside* that lock, via a per-subscriber queue, so that a slow or unresponsive subscriber cannot delay other requests.

Errors are reported with the gRPC status code appropriate to each condition rather than being silently ignored: `NOT_FOUND` for an unknown restaurant, an unavailable item or an unknown order; `PERMISSION_DENIED` when a restaurant attempts to modify an order belonging to another; and `FAILED_PRECONDITION` for a transition the state machine disallows, including cancellation of an order already accepted.

5.4 Verification

The system was exercised with the server on one compute node and clients driven from two others. The sequence demonstrated was: retrieval of both restaurant menus; placement of an order for one Margherita Pizza and two Garlic Breads, for which the server computed the expected total of 550 and the initial status `PLACED`; the restaurant advancing the order through `ACCEPTED`, `PREPARING` and `READY`, with each transition delivered to the subscribed customer over the stream in order; a second customer placing a concurrent order and receiving a distinct identifier; and each of the six error conditions above returning its intended status code.

5.5 Known limitation

`SubscribeToOrderUpdates` waits on its queue without a timeout while occupying a thread from a pool of ten. Ten customers tracking orders simultaneously — ordinary use — exhaust the pool, after which the server ceases to answer any request. Subscribing to an order that has already reached `READY` blocks indefinitely for the same reason, as no further update will ever be produced. Both follow from the same cause and are addressed by waiting on the queue with a timeout while re-checking whether the client remains connected, by transmitting the current status upon subscription, and by enlarging the thread pool.

6 Execution Environment and Method

Measurements were taken on the RCE cluster. Compute nodes are dual Intel Xeon Gold 5317 processors, providing 24 physical cores and 48 logical processors with hyperthread siblings at a stride of 24, with 502 GB of memory and a shared NFS home directory. Jobs are allocated by Slurm. Servers and clients are always placed on distinct compute nodes, never on the login node, and always addressed by node name rather than by `localhost`.

A methodological point affecting all timings. Requesting `--nodes=N --ntasks-per-node=1` without also requesting `--cpus-per-task` yields *one physical core per node*. The reported processor count of two is misleading, as those are the two hyperthreads of a single core: under such an allocation two mapper processes on one node require 2.49s where one requires 1.27s, and `mpirun -np 8` confines eight ranks to a single core. Every measurement in this report requests `--cpus-per-task` explicitly, so that N tasks and N MPI ranks each receive N physical cores. Without this precaution the MPI program appears not to scale at all, which is an artefact of the allocation rather than a property of the program.

Reproducibility. Datasets are produced by a generator with fixed seeds. Every performance run records its own correctness verdict alongside its timings, and runs that did not reproduce the reference output would be excluded from the analysis; no run in the reported sweeps failed to do so. The figures in this report are generated directly from the recorded result files.

7 Conclusions

The same problem in two paradigms. Sections 3 and 4 implement identical analytics as a batch job and as a streaming system. The batch implementation is simpler and produces its answer only at completion; the streaming implementation maintains the answer continuously and serves queries during ingestion, at the cost of a coordinator through which all data must pass. Neither is uniformly preferable: the choice follows from whether the answer is needed during ingestion or only after it.

State representation mattered more than the framework. The clearest result of the MPI comparison is that the difference in speed is attributable to dense versus sparse state, not to MPI versus MapReduce. The same decision that makes our implementation $2.8\times$ slower also makes it use $39\times$ less memory, and that reduction is what permits the work to be divided among mappers that each observe only part of the input.

Message granularity dominated the streaming system. Varying the batch size changed throughput by a factor of 296, far exceeding the effect of the worker count or the routing strategy. Where a cost is incurred per message rather than per record, aggregation into larger messages is the most effective optimisation available.

Each system’s limit was identified and attributed. The batch pipeline is bounded by parsing throughput and by the physical cores available; the streaming system is bounded by its single coordinator under concurrent query load; the ordering service is bounded by the thread pool consumed by long-lived subscriptions. In each case the constraint was located by measurement rather than inferred, and the corresponding remedy follows from the diagnosis.